

Architectural Analysis of High-Throughput Real Estate Data Systems and API Scalability

The global real estate technology landscape has undergone a fundamental transformation, transitioning from a fragmented collection of regional datasets into a sophisticated, interconnected ecosystem of high-velocity data streams. In this environment, scalability is no longer defined merely by the ability to handle a high volume of requests per second, but by the capacity to deliver normalized, multi-sourced property intelligence across diverse cloud architectures with minimal latency.¹ As organizations increasingly leverage artificial intelligence and machine learning to drive underwriting, risk assessment, and market analysis, the demand for "AI-ready" data—information that is pre-structured, standardized, and accessible via high-throughput delivery mechanisms—has become the primary benchmark for selecting a data partner.¹

The most scalable solutions in 2026 are characterized by a departure from traditional, rate-limited REST architectures toward cloud-native data sharing and bulk delivery platforms such as Snowflake, Amazon S3, and Databricks.⁶ This shift reflects a broader industrial requirement for "wide data" processing, where thousands of attributes across millions of parcels are analyzed simultaneously to identify market anomalies or investment opportunities.⁹ Providers like ATTOM, CoreLogic (now Cotality), and BatchData have emerged as the dominant forces in this space, each offering distinct architectural approaches to solving the "last mile" problem of real estate data integration.²

Architectural Paradigms of Modern Real Estate APIs

The scalability of a real estate API is intrinsically linked to its underlying architecture and the methodology used for data normalization. Large-scale enterprise users typically require nationwide coverage, which encompasses approximately 155 million to 160 million U.S. properties.¹ Managing this volume of data necessitates a robust ingestion engine capable of processing updates from over 3,200 unique sources, including county recorders, tax assessors, and fragmented Multiple Listing Services.¹⁰

The evolution of these platforms shows a clear trend toward unified identifiers. For instance, CoreLogic's Cotality Integrated Property (CLIP®) ID provides a persistent, machine-learning-driven record identifier that connects disparate datasets—such as tax, deed, and mortgage records—to a single property entity.⁴ This innovation is a critical component of scalability, as it eliminates the need for complex, resource-intensive string matching on the client side, allowing for seamless joins across massive datasets within a cloud environment.⁴

Core Enterprise Property Data Infrastructure Comparison

Feature	ATTOM Data	CoreLogic (Cotality)	BatchData	Cherre
Primary Focus	Nationwide Public Records	Mortgage/Risk/MLS	Multi-source Enrichment	Data Integration/Fabric
Property Coverage	158M+ Parcels	5.5B+ Records Global	155M+ U.S. Parcels	Unified Aggregator
Standardization	20-Step Quality Check	CLIP® ID System	USPS Address Validation	Schema Stitching
Cloud Native	Snowflake/Databricks	Databricks/GCP/Snowflake	Snowflake/S3/BigQuery	GraphQL/Snowflake
Update Cadence	Daily to Weekly	Real-time to Weekly	Daily/Continuous	Incremental/Real-time
Key Architectures	REST/Bulk/Cloud	OData/RESO Web API	REST/Bulk/S3	GraphQL
Data Width	9,000+ Attributes	94% MLS Coverage	700-1,000+ Points	100+ Petabytes

The transition toward cloud-native data sharing represents a pivotal shift in the proptech sector. While REST APIs remain the standard for real-time, single-property lookups, they often become a bottleneck for enterprise applications requiring bulk ingestion or complex analytical modeling.

The most scalable delivery method currently available is "Secure Data Sharing," exemplified by the partnerships between major data providers and Snowflake.⁴ In this model, the data does not need to be moved or copied via traditional Extract, Transform, Load (ETL) processes. Instead, the provider grants the consumer direct, secure access to the data residing in the provider's Snowflake instance.¹⁸ This architecture provides near-zero latency for bulk operations, as updates made by the provider are reflected in the consumer's environment in near real-time.¹⁹

CoreLogic and the OData Standard for MLS Scalability

CoreLogic, recently rebranded as Cotality, remains the foundational infrastructure for much of the North American real estate market.² Its Trestle platform serves as a primary distribution hub for Multiple Listing Service (MLS) data, utilizing the RESO Web API and RESO Data Dictionary standards to ensure interoperability across disparate regional systems.²⁰ The use of OData (Open Data Protocol) in Trestle allows for sophisticated, server-side filtering that is essential for scaling applications that must process millions of listing updates.²¹

The OData protocol facilitates a degree of query precision that traditional REST endpoints often lack. For instance, developers can use the \$select parameter to refine the fields returned in a response, reducing bandwidth consumption and improving client-side processing speeds.²¹ Furthermore, the \$expand parameter enables the retrieval of related resources—such as high-resolution media or room-level details—in a single request, minimizing the "n+1 query" problem that plagues less sophisticated architectures.²⁰

Trestle API Technical Protocols and Capabilities

Protocol/Parameter	Mechanism	Scalability Advantage
OAuth2 Flow	Client Credentials	Standardized, Secure Auth ²¹
\$filter	Server-side Boolean Logic	Reduces Data Volume ²³
\$expand	Linked Resource Ingestion	Minimizes API Round-trips ²⁰
Replication=True	Sync Optimization	High-throughput DB Sync ²¹
Matrix Direct	Native Field Access	Unaltered Source Fidelity ²²
CLIP® ID	ML-driven Mapping	Eliminates String-match Latency ⁴

The technical depth of the CoreLogic ecosystem is particularly evident in its handling of high-volume data synchronization. The "Replication" endpoint in the Matrix database architecture is designed specifically for syncing entire databases or large subsets, appending /replication to the request path to optimize server-side indexing for rapid data transfer.²⁴ This is a critical feature for national aggregators and institutional analysts who must maintain local replicas of regional MLS data for complex time-series analysis or market forecasting.

ATTOM Data Solutions and the Enterprise Data Warehouse

ATTOM Data Solutions provides a nationwide property data warehouse that is widely regarded as one of the most comprehensive resources for public record information in the United States.¹ Its API architecture is engineered for enterprise-grade reliability, offering 9,000 unique property attributes across 158 million parcels.¹ The scalability of ATTOM's solution is bolstered by its "cloud-native" delivery options, which allow institutional users to query data directly within Snowflake or Databricks.³

For real-time applications, ATTOM provides a suite of RESTful endpoints that deliver property details, tax assessments, and automated valuation models (AVMs) with an average query latency of 0.4 to 0.5 seconds.²⁶ This performance is sustained even under high load through a robust rate-limiting strategy that typically enforces a cap of 200 calls per minute for standard enterprise accounts, with the flexibility to adjust these limits based on specific response time requirements and production needs.²⁷

ATTOM API Throughput and Rate Limiting Architecture

Limit Type	Reset Interval	Standard Enterprise Cap
Minute Counts	Second #0 of every minute	200 calls / min ²⁹
Hourly Counts	Minute #0 of every hour	Tier-dependent ²⁹
Daily Counts	Midnight UTC	Tier-dependent ²⁹
Monthly Limits	End of calendar month UTC	1M+ calls base ²⁹

²⁷

The structural integrity of ATTOM's data is maintained through a 20-step standardization process, which ensures that disparate data points from thousands of counties are normalized

into a consistent schema.² This normalization is the prerequisite for "AI-ready" data; without it, machine learning models trained on nationwide datasets would be prone to significant errors due to regional variations in data reporting. By resolving these inconsistencies at the ingestion layer, ATTOM allows developers to focus on building sophisticated logic rather than managing fragile ETL pipelines for fragmented public records.⁵

Modern High-Velocity Data: BatchData and the AI Paradigm

BatchData has emerged as a high-growth alternative to legacy providers, particularly for applications that require daily updates and integrated contact enrichment.¹⁰ Its platform covers 155 million U.S. properties and homeowners, offering between 700 and 1,000 data points per record.¹⁰ BatchData’s architecture is designed for modern proptech workflows, providing low-latency REST endpoints alongside bulk delivery options via AWS S3 and Snowflake.⁸

A significant differentiator for BatchData is its focus on owner intelligence and "skip tracing" capabilities. The platform boasts a 76% accuracy rate for "Right Party Contacts," which is significantly higher than industry averages, and it provides automated entity resolution to uncover the true owners behind corporate veils (LLCs).¹¹ This level of enrichment is vital for lead generation and outreach platforms that scale by automating the identification of motivated sellers or distressed property owners.¹⁰

BatchData API Pricing and Scale Optimization

Plan Tier	Monthly Investment	Record/Request Volume	Overage Cost
Growth	\$1,000	100,000 records	\$0.01 / call ³³
Professional	\$2,500	300,000 records	Volume-adjusted ³³
Scale	\$5,000	750,000 records	\$0.0066 / call ³³
Enterprise	Custom	Millions+	Negotiated ¹⁰

¹⁰

The economic scalability of the BatchData model is a primary attraction for startups and high-growth firms. By offering tiered plans that reduce the per-record cost as volume increases, BatchData enables businesses to scale their data consumption linearly with their growth.³² Furthermore, their modular add-on system allows organizations to pay only for the specific

datasets they require—such as building permits for \$1,250 per month or contact enrichment for \$5,000 per month—ensuring that architectural complexity does not lead to unnecessary expenditure.¹⁰

GraphQL and the Data Fabric Architecture

While OData and REST serve many enterprise needs, the emergence of GraphQL as a "data fabric" solution has redefined the upper limits of scalability for complex, multi-source integrations. Cherre represents the pinnacle of this approach, utilizing a single GraphQL API to connect thousands of internal and external datasets into a unified source of truth.⁹ This architecture is particularly effective for "wide and slow" data scenarios, where an application needs to filter on any of thousands of columns across hundreds of tables without needing to understand the underlying join logic.⁹

GraphQL's primary advantage in real estate is its ability to eliminate over-fetching and under-fetching. Clients can query only the specific objects and fields they require, and the server-side logic handles the complex relationships—one-to-many, many-to-many—between parcels, deeds, mortgages, and neighborhood analytics.⁹ For example, a developer can standardise an address using a single GraphQL query, which Cherre resolves against its 3.3 billion addresses using machine learning algorithms.¹⁷

Comparison of Real Estate Data Architectures

Architecture	Best Use Case	Key Mechanism
RESTful	Real-time lookup, mobile apps	Resource-based endpoints ¹¹
OData	MLS sync, complex filtering	URL-driven query logic ²¹
GraphQL	Unified data fabric, AI training	Schema stitching, custom fields ⁹
Cloud Share	Institutional analytics, big data	Direct storage access (Snowflake) ⁴
Webhooks	Event-driven triggers, alerts	Push notifications via HTTP POST ³⁷

4

The implementation of "spatial queries" is another hallmark of the Cherre GraphQL approach.

Using native PostGIS functions via the Hasura framework, Cherre allows clients to pull data based on custom-drawn geographical areas (polygons) without the performance degradation typically associated with geospatial computations in a standard REST environment.³⁴ This capability is indispensable for developers building map-centric applications that require real-time analysis of property densities or market trends within specific neighborhood boundaries.

Scaling Implementation: Rate Limiting and Traffic Management

As real-world APIs encounter increasing pressure from AI agents and automated scraping tools, the management of global state and rate limits becomes a critical infrastructure concern. Sophisticated API gateways—such as Zuplo, Apigee, and Kong—are employed by real estate data providers to ensure fair usage and maintain system stability.³⁹

Zuplo offers programmable rate limiting using TypeScript, which is particularly effective for multi-tenant APIs that need to apply different limits based on user tiers or specific API keys.³⁹ This is contrasted with Apigee (Google), which utilizes enterprise XML policies like "SpikeArrest" to smooth traffic by converting a per-minute rate into smaller, sub-second intervals.³⁹ For a provider like Bridge Interactive, this results in "Burst Rate Limits" that communicate available capacity via specific HTTP headers, such as Burst-RateLimit-Remaining and Burst-RateLimit-Reset.⁴²

Rate Limiting Algorithm Efficiency

Algorithm	Performance Logic	Best For
Token Bucket	Constant refill rate with burst allowance	Handling legitimate bursts ³⁹
Leaky Bucket	Constant processing rate, smoothing peaks	System stability, smoothing spikes ³⁹
Sliding Window	Continuous tracking over moving window	Precise, accurate rate limiting ³⁹
Fixed Window	Resets at start of each time block	Simple cases, low precision ³⁹

³⁹

Enterprise users must design their systems to handle the inevitable "429 Too Many Requests"

response code. Robust implementation involves checking the Retry-After header and utilizing exponential backoff or background queuing to re-process requests once the limit has reset.⁴¹ Some providers, like Attio, even utilize "score-based" rate limits, where the complexity of a query—determined by the number of filters, sorts, and expanded records—contributes to a cumulative score that determines when a user is throttled.⁴³ This incentivizes developers to write efficient queries and prevents a single complex request from monopolizing server resources.

Scalability through Event-Driven Architectures and Webhooks

To achieve true real-time synchronization without the inefficiencies of constant polling, the most scalable systems utilize webhooks to push data changes to consumer endpoints.³⁸ Webhooks function as "reverse APIs," initiating communication from the data provider to the client application the moment a specific event occurs, such as a price drop, a new listing, or a change in ownership.³⁸

Fluximmo, for instance, provides real-time real estate adverts through webhooks, ensuring that clients gain access to new listings within moments of their appearance on the market.³⁸ This event-driven model is particularly effective for property "Alerts," where the provider only transmits data that matches the client's specific search criteria.³⁸ For property data—which is more dynamic and prone to merging and de-duplication—the approach is often to send a list of unique identifiers (IDs) that the client can then fetch via the REST API, striking a balance between real-time awareness and data consistency.³⁸

Webhook Implementation and Security Best Practices

Implementation Step	Technical Requirement	Importance for Scalability
Endpoint Creation	HTTP POST handler (Express/Flask)	Receives the JSON payload ³⁸
Verification	HMAC-SHA256 Signature headers	Prevents unauthorized payloads ⁴⁵
Retry Policy	Exponential backoff (e.g., 2s, 4s, 8s)	Ensures delivery during downtime ³⁷
Timestamp Checks	Verification of window (e.g., 5 min)	Prevents replay attacks ⁴⁸

Deduplication	Unique Event ID tracking	Prevents processing the same change twice ⁴⁵
----------------------	--------------------------	---

37

The use of webhooks is becoming a prerequisite for "social" or "interactive" real estate environments. Amity, for example, leverages webhooks to transform social events—such as a user modifying their profile or sharing a property—into push notifications that are delivered to other users in real-time.⁴⁷ This model allows the application to remain passive, without requiring cron jobs or frequent API calls, thus reducing the infrastructure load for both the provider and the consumer.³⁸

High-Throughput Cloud Delivery: Snowflake and the AI Data Cloud

For organizations operating at a truly institutional scale—such as national lenders, insurance firms, and major proptech platforms—the most scalable architecture is often one that completely bypasses individual API calls in favor of massive-scale cloud ingestion.³ Snowflake's "AI Data Cloud" has become the de facto standard for this type of delivery, enabling data providers like ATTOM, Cotality, and HousingWire to share their entire datasets with clients through secure, cross-cloud connectivity.⁴

The primary scalability driver in this context is the elimination of ETL overhead. By using tools like Snowpipe for continuous ingestion or the COPY INTO command for bulk loading from S3 stages, enterprises can ingest terabytes of raw transactional data into structured tables within minutes.⁷ Snowflake's ability to isolate warehouses—providing dedicated compute resources for ingestion that do not interfere with analytical queries—ensures that the system remains fault-tolerant and performant even during peak loading periods.⁴⁹

Performance KPIs for Snowflake Real Estate Workloads

Metric	Definition	Enterprise Target
Data Freshness	Latency from source to Snowflake query	< 5 minutes for transactions ⁵²
Query Latency	Time for dashboard results to return	P90 < 3 seconds ⁵²
Throughput	Volume of data processed per window	50 GB / 30 minutes ⁵²

Concurrency	Simultaneous users without degradation	100+ concurrent peak ⁵²
Cost Efficiency	Credit consumption vs business value	Defined per-job credit budget ⁵²

49

A critical consideration for technical architects is the management of "architectural debt" within these cloud environments. Snowflake's inherent simplicity can lead to the implementation of inefficient queries that consume excessive credits.⁵² Achieving economic scalability requires active monitoring of COPY_HISTORY and SYSTEM\$PIPE_STATUS to identify bottlenecks, as well as the strategic use of file sizes (preferably 100 MB to 250 MB) and path partitioning to minimize unnecessary API calls and compute usage during the ingestion process.⁴⁹

Niche Data Layers: Risk, STR, and AI-Powered Evaluations

Beyond the core property records, true enterprise scalability often depends on the integration of highly specialized datasets that provide qualitative context to the quantitative records.

Specialty Real Estate Analytics and Evaluation APIs

- **Property Evaluation and Computer Vision:** Homesage.ai is unique in its use of advanced Computer Vision AI to analyze property images for evaluation accuracy.³⁶ By extracting data on renovation quality and architectural style from visual inputs, it provides a layer of intelligence that traditional AVMs lack, supporting more nuanced investment decisions.³⁶
- **Climate Exposure and ESG Scoring:** ClimateCheck and similar APIs assess property-level risks for flood, wildfire, drought, and heat.¹ This data is increasingly integrated into underwriting systems and investor dashboards to support risk-informed decision-making in a changing environment.¹
- **Short-Term Rental (STR) Performance:** AirDNA and Mashvisor provide metrics on occupancy rates, average daily rates (ADR), and revenue projections for vacation rentals.¹ This allows investors to compare STR vs. long-term rental (LTR) strategies across 11 million active listings.⁵⁴
- **Multifamily and Pipeline Analytics:** Smart Apartment Data delivers extensive multifamily analytics across 230,000 properties, including details on 14,000 pipeline projects and 11,000 management firms.⁵⁵ This granular level of detail is essential for suppliers and locators operating in the multifamily sector.⁵⁵

These specialized APIs are often used in combination with foundational property data APIs like

ATTOM or CoreLogic to create a "stacked" data architecture.⁵⁶ For example, a property management platform might pull core characteristics from ATTOM, valuation forecasts from HouseCanary, and resident support via AI chatbots like STAN AI, which are integrated directly into the property management workflow.³⁶

The Future of Real Estate Intelligence: AI Agents and Conversational Data

As the volume of available data increases, the bottleneck has shifted from "data access" to "data synthesis." The next frontier of real estate scalability is the development of AI agents capable of querying and processing this vast data ocean using natural language.⁵⁷

AppFolio’s Realm-X represents a significant milestone in this evolution, embedding a generative AI conversational interface throughout the property management workflow.⁵⁷ This assistant functions as a co-pilot for managers, generating reports, managing vendor communications, and drafting resident responses in seconds.⁵⁷ Similarly, Prophia uses a hybrid AI approach to extract data from commercial leases, maintaining one of the largest private commercial real estate datasets to train its models on billions of lease provisions.⁵⁷

AI and Machine Learning Integration Benchmarks

AI Application	Mechanism	Industry Impact
Lease Abstraction	OCR + ML Extraction	High-impact operational efficiency ⁵⁷
Predictive Maintenance	Anomaly detection algorithms	Reduces equipment breakdown costs ⁵⁷
Conversational Search	LLM + Vector DB Querying	Simplifies complex data exploration ⁵⁷
Automated Underwriting	AI data aggregation + comps	Accelerates loan approval windows ⁵⁷

⁵⁷

For technical teams, the scalability of these AI agents is intrinsically linked to the underlying data architecture. APIs that are designed with AI agent consumption in mind—featuring role-based access controls, low-latency database connectors, and structured data payloads—will become the foundation of the 2026 proptech stack.⁵⁹ Currently, only 24% of developers actively design APIs for AI agent consumption, suggesting a significant opportunity

for competitive advantage for those who prioritize "agent-readiness" in their infrastructure.⁵⁹

Economic Realities: The Cost of Nationwide Scalability

While technical scalability is paramount, it cannot be separated from the economic realities of data procurement. The market is dominated by a few "legacy" providers who have historically acquired many smaller competitors, leading to a landscape where high-volume nationwide data is both premium-priced and essential.²⁸

Real Estate Data Provider Pricing and Market Position

Provider	Target Market	Pricing Philosophy	Transparency
ATTOM	Institutional/Enterprise	Custom, Volume-based	Opaque/Negotiated ²⁸
CoreLogic	Financial/Mortgage	Value-based / Professional	Opaque/Tailored ⁶¹
BatchData	Tech-forward Startups	Tiered Monthly Plans	Transparent ³³
Zillow	Consumer/Light App	Free / Restricted Comm	High ¹¹
Mashvisor	Investors/Analysts	Subscription-based	Medium ¹¹
Estated	Digital Teams	Predictable, Annual	Medium ²⁸

¹¹

For national aggregators like DealCheck or RentCast, the cost of licensing data from a primary provider can exceed \$10,000 per month for basic nationwide sets at larger volumes.⁶⁰ This cost pressure has led some larger players to source their own property data in-house or to utilize hybrid models that combine official APIs with sophisticated scraping tools—like ScrapingBee—to unify listings from sources that lack a formal API.¹²

The most cost-effective path to scalability often involves a "land and expand" strategy. Organizations start with a lower-tier plan (e.g., BatchData's \$1,000/month "Growth" plan) to

validate their workflow and scale to custom enterprise agreements only once their user base or query volume justifies the investment.³³ For institutional firms, the use of Snowflake marketplaces can actually reduce total cost of ownership by eliminating the labor and infrastructure costs associated with managing a multi-terabyte data warehouse.¹⁹

Strategic Synthesis on API Scalability

Identifying the "most scalable" API for real estate data requires a nuanced understanding of the specific use case. No single provider is superior in every dimension of scalability—throughput, data width, delivery variety, and economic flexibility.

Synthesis of Provider Superiority by Scalability Dimension

1. **Architectural Versatility (The Hybrid Winner): ATTOM Data.** By offering a robust REST API, bulk file transfers, and cloud-native Snowflake/Databricks delivery, ATTOM provides the most comprehensive array of scaling paths for a single nationwide dataset.¹
2. **Data Ingestion and Standardisation (The Interoperability Winner): CoreLogic (Cotality).** Through its Trestle platform and CLIP® ID system, CoreLogic provides the most scalable way to ingest and normalize fragmented MLS data at a national level, adhering to the latest RESO and OData standards.⁴
3. **Modern Developer Experience (The Growth Winner): BatchData.** For high-velocity startups requiring daily updates, integrated skip tracing, and transparent volume-based pricing, BatchData offers the most scalable developer-centric platform with the lowest friction for rapid deployment.¹¹
4. **Complex Data Aggregation (The Knowledge Graph Winner): Cherre.** For enterprises that must integrate internal, public, and dozens of third-party datasets into a single, queryable source, Cherre's GraphQL data fabric is the most scalable solution for managing "wide" data at an institutional scale.⁹

The selection of a data partner in 2026 should be guided by a "future-proof" architecture. For organizations building on-demand mobile applications, the latency profiles and rate limits of a provider like ATTOM or RealEstateAPI.com are the primary metrics. For those building deep-learning models or institutional portfolio analytics, the zero-ETL capabilities of a Snowflake-integrated provider like HousingWire or Cotality are paramount. Finally, for those building the next generation of AI-driven lead generation and outreach platforms, the data enrichment depth and entity resolution accuracy of BatchData provide the necessary foundation for sustainable commercial growth. The most scalable API is not a single product, but an ecosystem that aligns its data delivery and economic models with the technical maturity of the consuming organization.

Works cited

1. The 10 Best Real Estate APIs in 2026 - Attom Data, accessed April 28, 2026, <https://www.attomdata.com/news/attom-insights/best-apis-real-estate/>

2. 6 Best Real Estate Data Providers In 2026 (Expert-Rated) - Homesage.ai, accessed April 28, 2026, <https://homesage.ai/resources/blog/best-real-estate-data-providers-2026/>
3. How ATTOM Is Bridging the Gap Between Property Data and Sophisticated AI Workflows, accessed April 28, 2026, <https://www.attomdata.com/news/company-news/data-solutions/how-attom-is-bridging-the-gap-between-property-data-and-sophisticated-ai-workflows/>
4. Cotality Property Data & CLIP App on Snowflake Marketplace, accessed April 28, 2026, <https://www.cotality.com/our-data/data-marketplaces/snowflake>
5. RealEstateAPI | Modern Real Estate Data API for Developers, accessed April 28, 2026, <https://www.realestateapi.com/>
6. Export Results From a Snowflake Clean Room to an AWS S3 Bucket, accessed April 28, 2026, <https://docs.liveramp.com/connect/en/export-results-from-a-snowflake-clean-room-to-an-aws-s3-bucket.html>
7. Bulk loading from Amazon S3 - Snowflake Documentation, accessed April 28, 2026, <https://docs.snowflake.com/en/user-guide/data-load-s3>
8. Property Data APIs Comparison — 2026 Guide - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/best-property-data-apis-developers-2026-comparison>
9. How Cherre creates real estate data products with Cloud SQL | Google Cloud Blog, accessed April 28, 2026, <https://cloud.google.com/blog/products/databases/how-cherre-creates-real-estate-data-products-with-cloud-sql>
10. Property Data API: Ultimate Selection Guide - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/ultimate-guide-choosing-property-data-api>
11. Top 7 APIs for Real Estate Data Enrichment - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/apis-real-estate-data-enrichment>
12. Best Real Estate APIs for Developers in 2026 - ScrapingBee, accessed April 28, 2026, <https://www.scrapingbee.com/blog/best-real-estate-apis-for-developers/>
13. Trestle Documentation: Trestle, accessed April 28, 2026, <https://trestle-documentation.corelogic.com/>
14. Top 10 Property Data Providers for Real Estate Investors - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/top-property-data-providers-real-estate-investors>
15. Property & Real Estate API – How It Works - Attom Data, accessed April 28, 2026, <https://www.attomdata.com/solutions/property-data-api/how-it-works/>
16. Best Real Estate Data Providers for 2026 - Attom Data, accessed April 28, 2026, <https://www.attomdata.com/news/company-news/data-solutions/best-real-estate-data-providers-for-2026/>
17. Real Estate Data Management and Insights Platform - Cherre, accessed April 28, 2026, <https://cherre.com/products/platform/>
18. Snowflake Marketplace for Providers, accessed April 28, 2026, <https://www.snowflake.com/en/product/features/marketplace/snowflake-marketplace-for-providers/>

19. ATTOM Snowflake Cloud Delivery - FAQs, accessed April 28, 2026, <https://www.attomdata.com/solutions/cloud-delivery/snowflake/faqs/>
20. Bridge API | Normalized Data for Developers | Bridge Interactive, accessed April 28, 2026, <https://www.bridgeinteractive.com/developers/bridge-api/>
21. Getting Started – Getting Started with Trestle WebAPI, accessed April 28, 2026, <https://trestle-documentation.corelogic.com/webapi.html>
22. Streamline Data Management Solutions with Trestle - Cotality, accessed April 28, 2026, <https://www.cotality.com/products/trestle>
23. CoreLogic Trestle (generic) Archives | RESO - Real Estate Standards Organization, accessed April 28, 2026, <https://www.reso.org/web-api-examples/mls/corelogic-trestle-generic/>
24. Direct Web API - Listing (MLO Data) Reference - Trestle Documentation, accessed April 28, 2026, <https://trestle-documentation.corelogic.com/direct-webapi-mlo-reference.html>
25. Postman - miami realtors, accessed April 28, 2026, <https://www.miamirealtors.com/wp-content/uploads/bsk-pdf-manager/2019/12/Trestle-Docummentation-Web-API-Registration-Steps.pdf>
26. ATTOM API Documentation & Examples, accessed April 28, 2026, <https://api.developer.attomdata.com/home>
27. Top APIs for Historical Property Data Integration - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/top-apis-for-historical-property-data-integration>
28. How APIs Are Transforming Real Estate: Trends, Use Cases, and ..., accessed April 28, 2026, <https://oril.co/blog/the-real-estate-api-transformation-how-data-driven-innovation-is-reshaping-the-industry/>
29. When does my API usage get reset? - ATTOM Cloud Help, accessed April 28, 2026, <https://cloud-help.attomdata.com/article/631-api-usage-reset>
30. Property Data Solutions for Your Business - BatchData API, accessed April 28, 2026, <https://batchdata.io/api-solutions>
31. BatchData: Real Estate Data, APIs, and AI Solutions, accessed April 28, 2026, <https://batchdata.io/>
32. Mastering Your Business with a Real Estate Data API - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/real-estate-data-api>
33. Real Estate Data API Pricing Compared - BatchData, accessed April 28, 2026, <https://batchdata.io/blog/real-estate-data-api-pricing-comparison-batchdata-compettitors>
34. Case Study: Cherre's data accessible by a single GraphQL API - Hasura, accessed April 28, 2026, <https://hasura.io/customers/cherre>
35. WeBuilt - Cherre API integration - HackMD, accessed April 28, 2026, <https://hackmd.io/D2Ud3QimTtCxIau4Oyfvjg>
36. 8 Best Property Evaluation APIs In 2026 (AVM Alternatives Compared) - Homesage.ai, accessed April 28, 2026, <https://homesage.ai/resources/blog/8-best-property-evaluation-apis-in-2026/>
37. Notifications - Atto Docs, accessed April 28, 2026, <https://docs.atto.co/content/docs/guides/platform-integration/notifications>

38. Webhooks - Real-estate data API - Fluximmo V2 (2.1.0) - Authentication, accessed April 28, 2026, <https://doc.fluximmo.io/doc/main-concepts/webhooks>
39. API Rate Limiting Comparison: Which Platforms Have the Best Built-In Features? - Zuplo, accessed April 28, 2026, <https://zuplo.com/learning-center/api-rate-limiting-platform-comparison>
40. Rate Limiting in Multi-Tenant APIs: Key Strategies - Blog, accessed April 28, 2026, <https://blog.dreamfactory.com/rate-limiting-in-multi-tenant-apis-key-strategies>
41. What is API Rate Limiting? Understanding Request Throttling and Best Practices, accessed April 28, 2026, <https://blog.postman.com/what-is-api-rate-limiting/>
42. New per-minute burst rate limits coming soon - Bridge Interactive, accessed April 28, 2026, <https://www.bridgeinteractive.com/new-per-minute-burst-rate-limits-coming-soon/>
43. Handling rate limits - Attio Docs, accessed April 28, 2026, <https://docs.attio.com/rest-api/guides/rate-limiting>
44. Article: Boomi Enterprise Platform API Rate of Requests per Second, accessed April 28, 2026, <https://community.boomi.com/s/article/atomsphereapirateofrequestspersecond>
45. Webhooks(Real-time updates) - API Documentation, accessed April 28, 2026, <https://developers.pipelinersales.com/api-docs/core-api-concepts/webhooks-real-time-updates>
46. Subscribe to Push Notifications | YouTube Data API - Google for Developers, accessed April 28, 2026, https://developers.google.com/youtube/v3/guides/push_notifications
47. Transforming Webhooks into Push Notifications: A Step-by-Step Guide | by Italo Orihuela, accessed April 28, 2026, https://medium.com/@italo_2018/transforming-webhooks-into-push-notifications-a-step-by-step-guide-925f4591ef13
48. Webhooks - Fathom API, accessed April 28, 2026, <https://developers.fathom.ai/webhooks>
49. Best Practices for Data Ingestion with Snowflake - Part 1, accessed April 28, 2026, <https://www.snowflake.com/en/blog/best-practices-for-data-ingestion/>
50. HousingWire Data: Real Estate Sales Data - Property Transactions & Closed Sale Intelligence Platform - Snowflake, accessed April 28, 2026, <https://app.snowflake.com/marketplace/listing/GZTYZ43T813/housingwire-data-real-estate-sales-data-property-transactions-closed-sale-intelligence-platform>
51. Transfer Data from Amazon S3 to Snowflake - Braze, accessed April 28, 2026, https://www.braze.com/docs/user_guide/data/distribution/braze_currents/use_cases/s3_to_snowflake/
52. Performance - Snowflake, accessed April 28, 2026, <https://www.snowflake.com/en/developers/guides/well-architected-framework-performance/>
53. Snowflake Pricing | Choose the Right Edition for Your Data Needs, accessed April 28, 2026, <https://www.snowflake.com/en/pricing-options/>
54. Best Real Estate Data API in 2025: Mashvisor vs AirDNA vs Zillow, accessed April 28, 2026, <https://www.mashvisor.com/blog/best-real-estate-data-api-comparison/>

55. Top Real Estate Data Providers for Snowflake in 2026 - Slashdot, accessed April 28, 2026, <https://slashdot.org/software/real-estate-data/for-snowflake/>
56. Reliable real estate data providers with full property and neighborhood insights? - Reddit, accessed April 28, 2026, https://www.reddit.com/r/RealEstateTechnology/comments/1oh84x9/reliable_real_estate_data_providers_with_full/
57. The Best AI Tools for Real Estate: A 2026 Field Guide - V7 Labs, accessed April 28, 2026, <https://www.v7labs.com/blog/best-ai-tools-for-real-estate>
58. Providers | Snowflake Marketplace, accessed April 28, 2026, <https://app.snowflake.com/marketplace/providers>
59. 38 REST API Performance Statistics Every Enterprise Should Know in 2026 - Dreamfactory, accessed April 28, 2026, <https://www.dreamfactory.com/hub/rest-api-performance-statistics>
60. What is the price of Attom data? : r/RealEstateTechnology - Reddit, accessed April 28, 2026, https://www.reddit.com/r/RealEstateTechnology/comments/1b76bfz/what_is_the_price_of_attom_data/
61. Top 10 Real Estate Data Providers in 2026 - PropertyShark, accessed April 28, 2026, <https://www.propertyshark.com/Real-Estate-Reports/top-10-real-estate-data-providers/>
62. Best Commercial Real Estate Data Sources for 2026 - CRE Daily, accessed April 28, 2026, <https://www.credaily.com/reviews/best-commercial-real-estate-data-sources/>
63. Property Data for Real Estate Insights - Datafiniti, accessed April 28, 2026, <https://www.datafiniti.co/data/property-data>